Fastronomic ‘Alpha’ vs ‘Omega’- Product Technical Analysis and Optimisation Report

Name: Michal Kosnik

Library Card Number: 38802139

Executive Summary

Fastronomic is a company who produces sports meal replacements. They are part of an industry where there is growing demand for analyzing data using advanced modelling methodologies. This is because it can help guide their production and market strategy- especially in the context of their new high-protein product, Fastronomic Omega. The company is ready to migrate their systems from Excel, and utilizing Python, we translated the company's existing Excel-based model to optimise the blend for their main product, Fastronomic Alpha. Our model closely reproduced their blend, with the variance found to be within acceptable tolerance levels.

Like most sports powders/ meal replacements, Fastronomic Omega is created from a blend of base powder, protein, carbohydrates, and oil. Unlike Alpha, this one also includes the newly sourced P+-protein. This optimised blend enables Fastronomic to effectively address the dietary needs of athletes and those on high-protein diets.

To aid in production planning, a computer model was designed to maximise profit by optimising the volume of 2kg packs of Fastronomic Alpha and Omega needed to produce. This took into consideration many variables such as the cost of ingredients, quantities used of each ingredient, and the selling price of the products. Our model highlighted opportunities for cost-efficient production expansion, identifying specific ingredients which, if sourced in higher quantities, could lead to increased production levels and profit.

The potential demand for the new Omega product was also considered by comparing a current possible optimal production plan, against a plan that incorporated demand projections for the next cycle. This comparative analysis provides insights into the potential market response to the new product and helps to devise a balanced production strategy.

In summary, our Python-based modelling approach provides an efficient and easily generalisable tool for Fastronomic, and is an upgrade from their current system. The company is now better equipped to control costs, manage production, and plan for market demand. As a recommendation, Fastronomic should continue refining these models as new data and market feedback become available to ensure optimal

production planning and product mix. All Python programs are attached, and some extra snippets of code are in the appendix section of this report.

Methodology and Process

1. First Task Overview (Alpha Blend)

Fastronomic already had a model to determine the blend of powders for their Alpha product, which was created and calculated using Excel. Their first request was to convert this model to Python, which is a powerful and robust programming language that allows for more technical flexibility, as well as the ability to create a user-friendly interface, which isn't possible on Excel. Because of this, and the global issue of 'big data' and the increasing demand for its' analysis in business, Python is becoming widely adopted in many industries.

The conversion was relatively simple, and for the analysis a tool called 'PuLP' was used. This is an online library or 'plug-in' used to solve linear programming problems on Python. The results from this model aligned with Fastronomic's model, which verified that everything was accurate and working. The resulting proportions of ingredient powders (base, protein, carbohydrates, and oil) for the Alpha blend from the Python model were very similar to the results from their Excel model, with some very minor differences in the decimal points. I have been informed that these differences are expected and are within allowable tolerances, so are not a cause for concern.

1.1- Methodology

The first step in the process was to set up the linear programming (LP) problem. Most LPs are laid out in a similar template- create the mode, define the variables, set an objective function, add constraints, then solve and print. The objective function is what tells the program whether you want to maximise or minimise something when optimising.

In the case of Fastronomic Alpha, their objective is to minimise the total cost of the ingredients used per 125g portion. This is defined as the sum of the cost/gram of each individual ingredient * amount (in grams) of each ingredient used.

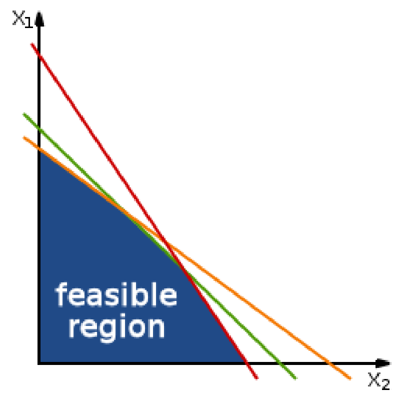
Next, we define the constraints for the problem. These constraints represent the nutritional requirements for the product, which include energy, fat, carbohydrate, protein, salt, and fibre. Each of these requirements was expressed as a linear inequality, with the total nutritional contribution from all ingredients being as close to the recommended daily amount. Additionally, the total weight of all ingredients used must equal the portion size of 125g.

Non-negativity constraints were also included, meaning that the amount of each ingredient used cannot be less than zero- which does actually happen in another model, discussed later. After defining the problem, we used the solve() function from the PuLP library to find the optimal solution.

1.2- Results and Evaluation

The solution provided by the solver included the optimal amounts of each ingredient to use in the Fastronomic Alpha product, as well as the total cost per portion. These results were found to be in close agreement with the amounts provided by the company, with minor discrepancies likely due to differences in rounding and numerical precision between Excel and Python:

Fastronomic Alpha ingredients:
Base (g) = 58.81
P-Protein (g) = 37.0
Carbohydrates (g) = 21.28
Oil (g) = 7.91



Graphical Representation of Linear Programming

One potential issue in this type of analysis is the presence of multiple optimal solutions- there may be an 'area' of 'feasible answers' to the problem. In Fastronomic's case, there may be several combinations of ingredients that satisfy all constraints and have the same total cost. This issue could be addressed by adding additional constraints or adjusting the objective function to differentiate between these solutions.

Another issue could be the sensitivity of the solution to changes in variables. If the cost of an ingredient were to change very slightly, it could impact the optimal blend of the product more than proportionately. Sensitivity analysis could be performed to understand the impact of such changes and inform strategic decisions around ingredient sourcing and pricing.

Lastly, theoretical solutions usually assume that all constraints must be strictly met or exceeded. In practice, there may be some flexibility or tolerance in these requirements, and it might be acceptable for some nutritional contributions to deviate slightly from target values. If this is the case, the constraints could be adjusted to reflect this flexibility, which might result in a lower cost solution.

Overall, the use of Python and the PuLP library proved to be an effective approach for solving this LP problem, providing insights that can inform cost-effective production decisions for the Fastronomic Alpha product.

2. Second Task Overview (Omega Blend)

Fastronomic is launching a new product called Fastronomic Omega, and they wanted to optimise the ingredient powder blend for this product. This product is meant to be an alternative rather than a replacement for Alpha, as it would be higher in protein and lower in carbohydrates, so it would cater to a different type of consumer such as athletes that have different dietary requirements. Another difference is that Omega uses a blend of five different powdered ingredients, unlike Alpha which uses only four. Omega contains a new 'P+' kind of protein powder- this has significantly more protein, as well as less carbs and salt.

They requested to create a similar optimised model for this new product, so they know how much of each ingredient to buy, as well as know how much they can expect to profit from Omega.

This was a similar process as the first analysis, where the task was to create a linear programming model in Python that creates a 125g blend of ingredients for the Omega product, which minimises price but keeps the nutritional value as close to RI as possible. However, there were some difficulties due to the increased number of variables, as well as some subjectivity about the constraints of this new product.

2.1 - Methodology

Methodology for this task was very familiar- the same template of the model was used, however it was adapted. There was an additional defined variable for the extra ingredient, and this model also solved for total price per portion to optimise more effectively; to compare to different products such as the Alpha powder; and as well as to record profits and revenues more accurately.

Mainly trial and error was used to find suitable constraints for this model, but as with any business analysis, manual critical and logical thinking is often needed in addition to technical analysis. Many drafts of the program gave outputs that were wildly incorrect, so the lower and upper bounds of certain constraints had to be adjusted, until a realistic answer was found. In fact, there were multiple 'correct' answers that were outputted which suggests there must be a feasible 'area', even if the whole area isn't optimised, just feasible. Although graphical programming was not used, this is a reasonable assumption to make when there are multiple solutions that fit within the criteria. Some examples of faulty outputs can be found in the appendix.

Finally, the code was cleaned up to run as efficiently and effectively as possible, and a model was found to be the most optimised. This finished program can also be found in the appendix.

2.2 - Results and Evaluation

This task proved more difficult for a couple of reasons. Because there were now five variables (ingredient powders) instead of four, this made the model more complicated and therefore less likely to solve a correct answer. The feasible region must have been nonexistent, given the current criteria. Also, the model had to balance the total price, weight of all the ingredients and the nutritional value- as a result a more advanced method could have been more suitable.

Additionally, the amount of each ingredient needed in the Omega blend was quite vaguely explained- a daily Reference Intake (RI) was provided, however this had some issues. This was a recommended amount of nutrition per day, and it is highly unlikely that anyone would be eating solely Fastronomic shakes, as this would defeat the health benefits- everyone should eat a full, healthy, mixed diet, especially athletes. This caused issues because there was no concrete value to set the constraints to in the Python model. Trial and error had to ultimately be used to calculate a viable option. Models like this can be very sensitive, especially with certain variables, so the first few iterations of the model produced very varied, unviable outputs- some of which can be found in the appendix section of this report. This must have meant that the constraints were too 'strict' and couldn't calculate a real answer.

However, Fastronomic mentioned that their Omega powder was to be high in protein and low in carbohydrates- this made it easier, as it meant the constraints could be relaxed so an optimised result could be calculated. The constraints for the other nutritional values were also sensibly adjusted to fit the needs of an athlete, while still staying within the daily RI.

Overall, it seemed that not all the ingredients were necessary- in nearly all iterations at least one ingredient was missing, and in some multiple had trace amounts of each ingredient. Technically this means that there are multiple 'solutions' to this model, and in practice it means that Fastronomic has flexibility when it comes to quantities of ingredients. However, they must still be careful, as some variables may be very sensitive, and small changes can deviate from the optimum. Further sensitivity analysis must be conducted to show exactly how much flexibility they have in their supply chain. The new P+ protein in particular, likely due to its' higher price. Most

models seemed to prefer the old protein to the new one- it's possible that a 50% increase in price wasn't worth the 40% increase in protein. Pricing will be mentioned in more detail in the following segment of this report.
Here is a comparison of the results, Alpha vs Omega (per 125g serving):

Alpha-nutrition	500.09 kcal	12.0g fat	40.0g carbs	25.0g protein	0.5g salt	9.05g fibre
Alpha-Ingredients	58.81g base	7.91g oil	21.28g carbs	37.0g protein	-	-
Omega-nutrition	500.0 kcal	16.40g fat	17.63g carbs	50.0g protein	0.36g salt	10.00g fibre
Omega-ingredients	0.0g base	24.04g oil	26.44g carbs	11.07g protein	63.45g P+ protein	

3. - Third task Overview (Product planning cycle)

This task was quite complicated and is known as a 'multidimensional knapsack problem'- meaning there are a lot of variables and interdependent groups of variables. The best approach to this was to first write down all the important data in logical order and calculate and simplify it manually, and then figure out a way to construct one cohesive model that focuses on just the production planning, with a fixed set of variables and values for ingredients and prices. Fastronomic did mention they are willing to adjust these numbers for this cycle and the next, and this will also be mentioned later in this report.

3.1 Methodology- Preparation

The aim of this task is to optimise their production planning- meaning, they want to calculate the optimum amount they should buy and sell to maximise profits. Therefore, costs and profits per unit had to be calculated so they could be cleanly integrated into this model. Since the previous two

models had all the necessary data, this could be worked out using relatively simple arithmetic.
Data had already been given for the Alpha ingredients (all converted from 125g to 2kg):

Alpha ingredient amounts per 2kg:
Base (g) = 940.96
P-Protein (g) = 592
Carbohydrates (g) = 340.48
Oil (g) = 126.56

Amounts for Omega had also been figured out using the model from task 2:

Omega ingredient amounts per 2kg:
Base (g) = 0
P-Protein (g) = 184.64
Carbohydrates (g) = 529.28
Oil (g) = 276.16
P+ Protein (g) = 1,009.76

Exact costs per pack were calculated by sections of code that solved total price and price per ingredient in a 125g serving, using the prices of the ingredients per kg. This was then multiplied by 16 to figure out how much a 2kg bag costs, then this was subtracted from the retail prices of the products. The results are as following:

Alpha cost of ingredients (per 2kg, in £):
Base = 6.56
P-Protein = 4.80
Carbohydrates = 1.44
Oil = 0.80
Total cost per 2kg bag = £13.60
Profit per 2kg bag = £11.56

Omega cost of ingredients (per 2kg, in £):
Base = 0.00
P-Protein = 1.44
Carbohydrates = 2.08
Oil = 1.60
P+ Protein = 12.16
Total cost per 2kg bag = £17.28
Profit per 2kg bag = £12.72

3.2 Methodology and Results

Using the quantities of ingredients, costs, and revenues for each product, we can now model an optimised production plan for Fastronomic's current situation. Further analysis of future scenarios will be reported later in this section. The model uses Alpha and Omega as the decision variables, and the available amounts of powder as constraints. According to the model, the optimum production plan is:

Optimal Production Plan:
Alpha = 29151.359 (2kg units)
Omega = 14852.662 (2kg units)
Maximum Profit: 525915.57068 (£)

This output from the program means that with their current means of production, they can produce around 29k packs of Alpha and 15k of Omega, to maximise their profit to over 500k.

Obviously not all the available materials would have been used up simultaneously, as each product uses varying amounts of each ingredient. Roughly multiplying the output amounts from this model and the known quantities of powders used per pack, it is strongly recommended Fastronomic purchase more protein. This is because they are by far the most used ingredient, as well as being one of the least available ones. If they purchase more protein, they'll be able to produce more- however they'd also have to purchase more of the other ingredients. They are already projected to have a nice profit of over half a million so they easily will have enough capital to invest/reinvest more in the future, whether it's into even more protein, or they can try experimenting with other ingredients, such as flavourings or other healthy additives- both of which could open them to a new market and increase their demand.

3.3 Conclusion and Recommendations

Especially for the Omega- since its' projected demand for the next cycle is meant to be only a fraction of the Alpha- likely due to the higher price and focus on athletes rather than casual consumers. Given the demand constraints, the model must be modified to include these too. The demand for the Alpha is at least 27,000 packs, which means they need to produce no less than that amount. On the other hand, the demand for the Omega is up to 8,000 packs, which means they should not produce more than that amount. It is recommended that these parameters can be added as constraints to the model if needed by Fastronomic. They could also bring more parameters in and create a more user-friendly interface, so they have the flexibility to create models easily, or use one big system to process all their data.

It would also be recommended to try to branch out their company into other sports foods/drinks- they are set for a very healthy profit with their current circumstances and have potential for even more. However, it seems like protein is a bottleneck for their growth- it is expensive, and their supplements are already saturated in it. if they want to truly replace people's meals, they have to be a lot healthier and more mixed, so perhaps they could use the gathered data and models to reevaluate their business operations.

1. Appendix

Some examples of infeasible outputs:

Optimal Blend for Omega:
B_base = 0.0
P_protein = 99.799599
C_carb = 100.2004
O_oil = 0.0
P+_protein = 0.0
Total cost per portion = 1.199198392
Total adds up to over 125, even though constraints were put in place at 125- these values appeared in a couple drafts of the model.

B = 2750.0g
Cost of B = £19.25
C = -1670.0g
Cost of C = £-6.68
O = 0.0g
Cost of O = £0.0
P = -880.0g
Cost of P = £-7.04
P_plus = 0.0g
Cost of P_plus = £0.0
Total cost = £5.53
This output happened one time when experimenting with printing price- for some reason extreme and negative values were output. It is very likely this was due to incorrect constraints, or incorrect objective function.